Homework 01

STAT 432

1. Least squares and matrix calculations

a. With seed 43201, the design matrix has dimension 100×6 and rank 6. I used the QR-based `lm.fit`, not an explicit inverse. The estimates are close to the generating coefficients, with ordinary sampling differences caused by the realized errors:

	Intercept	x_1	x_2	x_3	x_4	x_5
Truth	3.000	1.400	-0.900	0.700	0.000	1.100
Estimate	3.006	1.417	-0.888	0.795	0.048	1.132

b. From $\widehat{\mathbf{y}} = \mathbf{X}\widehat{\boldsymbol{\beta}}$ and $\mathbf{r} = \mathbf{y} - \widehat{\mathbf{y}}$, the training RMSE is

$$\sqrt{n^{-1}\mathbf{r}^{\mathsf{T}}\mathbf{r}} = 0.67637.$$

c. I obtain $\|\mathbf{X}^{\mathsf{T}}\mathbf{r}\|_\infty = 5.24 \times 10^{-14}$. The least-squares first-order condition is $-2\mathbf{X}^{\mathsf{T}}\mathbf{r} = \mathbf{0}$, so the nonzero value is only floating-point error. Thus the residual is orthogonal to every design column; because the intercept is a column, the residuals also sum to approximately zero.

2. Gaussian likelihood for linear regression

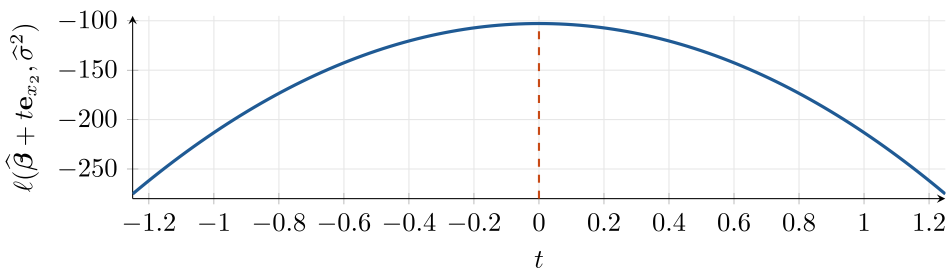
a. For fixed $\sigma^2 > 0$, the only term depending on $\boldsymbol{\beta}$ is $-\operatorname{RSS}(\boldsymbol{\beta})/(2\sigma^2)$. Therefore maximizing the likelihood is equivalent to minimizing RSS, and $\widehat{\boldsymbol{\beta}}_{\text{MLE}} = \widehat{\boldsymbol{\beta}}$.

b. Writing $v = \sigma^2$,

$$\frac{\partial \ell}{\partial v} = -\frac{n}{2v} + \frac{\text{RSS}}{2v^2} = 0 \quad \Longrightarrow \quad \widehat{\sigma}^2_{\text{MLE}} = \frac{\text{RSS}}{n} = 0.45748.$$

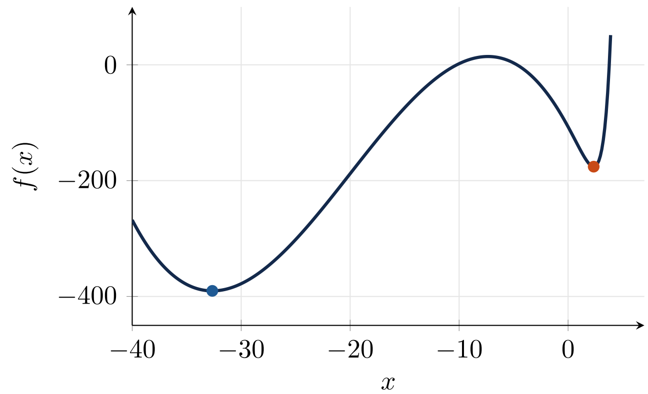
The second derivative at this solution is negative. The unbiased estimate is $\text{RSS}/(n-6) = 0.48668$. MLE divides by n because it directly optimizes the likelihood; unbiased estimation divides by $n-6$ because six fitted coefficients consume six degrees of freedom and $E(\text{RSS}) = (n-6)\sigma^2$.

c. The profile below uses $\widehat{\sigma}^2_{\text{MLE}}$ and changes only the x_2 coefficient. It is maximized at $t = 0$, exactly where the coefficient equals its least-squares estimate. This agrees with part (a): every nonzero t raises RSS and lowers the log-likelihood.



3. Starting values in nonconvex optimization

a. The plot shows two local minima separated by a local maximum, plus a sharp rise on the right. These features create separate basins of attraction for a local optimizer.



b. Supplying the stated derivative to BFGS gives:

Start	Final x	$f(x)$	$\lvert f'(x)\rvert$	Conv.
-15	-32.64911	-390.38577	8.73×10^{-8}	yes
0	2.34997	-175.86262	6.56×10^{-7}	yes

c. BFGS uses local function and gradient information, so the initial point determines which basin it reaches. The -15 run has the lower objective. A global claim would require ruling out all lower regions, for example by locating every root of f' and comparing all stationary values together with the limiting behavior as $x \to \pm\infty$. A dense grid or many starts is supporting evidence, but not by itself a proof.

4. Nearly collinear predictors

a. The augmented matrix is extremely ill-conditioned, even though predictive fit changes little:

Design	Condition number	Training RMSE
$\mathbf{X}$	1.525	0.67637
$\mathbf{X}_+$	20,209.204	0.67547

Response	$\widehat{\beta}_1$	$\widehat{\beta}_6$	Sum
$\mathbf{y}$	351.677	-350.259	1.418
$\mathbf{y}^\star$	341.677	-340.259	1.418

b. Changing the response from $\mathbf{y}$ to $\mathbf{y}^\star$ changes the x_1 coefficient by -10 and the x_6 coefficient by $+10$. Nevertheless, the two fitted vectors differ by only 0.001016 in RMSE and 0.002573 at their largest absolute difference.

c. Because $\mathbf{x}_6 - \mathbf{x}_1 = 10^{-4}\mathbf{u}$,

$$\mathbf{y}^\star - \mathbf{y} = 10^{-3}\mathbf{u} = 10(\mathbf{x}_6 - \mathbf{x}_1).$$

Hence coefficient changes $(-10, +10)$ reproduce the small response perturbation exactly in the augmented column space. Nearly canceling, individually huge coefficients can therefore yield stable fitted values. The data identify the combined contribution much better than the two separate effects, so separate causal or substantive interpretations would be unreliable.

5. Data preparation and summary statistics

a. The original data have 24 rows and 5 columns. `observation_id` and `group` are character; `x1`, `x2`, and `response` are numeric. There are 0 duplicated identifiers. Missing counts are 0 for every column except `response`, which has 3. After dropping only missing responses and adding `feature_sum=x1+x2`, the analysis table has 21 rows.

b. These summaries describe only the 21 observations with recorded responses:

Group	Observations	Mean response	Mean `feature_sum`
A	7	5.6000	3.6000
B	7	7.6029	4.9286
C	7	7.9971	6.5857

c. Sorting recorded responses from largest to smallest gives:

ID	Group	Response	feature_sum
obs-22	C	9.24	6.6
obs-16	B	9.00	5.2
obs-14	B	8.66	5.0

The code checks the retained-row count against the original nonmissing count, verifies no retained response is missing, and verifies identifier uniqueness (see the reproducibility section).

6. Can 39 million Fitbit records represent US adults?

a. The records are repeated days nested within roughly 59,000 people, not 39 million independent observations. Days from one person are correlated through persistent health, work, habits, and environment. Enrollment length, device-wearing and synchronization habits, technical access, illness, travel, device failure, and privacy choices can all change the number of contributed days. The naive daily-record mean gives a participant weight proportional to their valid recorded days, rather than equal participant weight; this matters if recording duration is related to activity.

b. No. BYOD versus WEAR was not randomized, so the 6,867 versus 5,797 median comparison does not identify a causal effect of prior Fitbit ownership. Existing owners may have greater income, more technology access, more health awareness, stronger exercise motivation, or different baseline health. The reported race and income differences directly show that the entry routes contain different populations, so confounding and self-selection are plausible explanations.

c. Selection occurs at several stages: voluntary All of Us enrollment; already owning a compatible device or being invited into WEAR; accepting, consenting, and connecting an account; and continuing to wear and synchronize the device on valid days. Each stage can depend on characteristics associated with activity, so observed participants and observed days need not represent US adults and their days. More records reduce random error for the selected participant-days but do not erase selection bias. The naive record-level mean also misses the stated equal-adult target by weighting prolific contributors more. A better analysis would first average within participant, then use population calibration/weighting when defensible, while accounting for within-person dependence and acknowledging unmeasured selection.

Source: Patten, T., Preble, E. A., Master, H., et al. (2026), "The All of Us Research Program's wearables dataset," *Nature Medicine*, 32, 2302–2310. https://doi.org/10.1038/s41591-026-04352-3.

Reproducibility code

The complete runnable script is also saved as `homework-01.R`. The compact listing below contains the calculations and required checks used for the report.

```
set.seed(43201); n <- 100; p <- 5
x <- matrix(rnorm(n*p), n, p)
colnames(x) <- paste0("x", 1:p)
X <- cbind(1, x)
beta <- c(3, 1.4, -.9, .7, 0, 1.1)
y <- as.vector(X %*% beta + rnorm(n, sd=.7))
fit <- lm.fit(X, y) # QR least squares
bhat <- fit$coefficients
r <- y - as.vector(X %*% bhat)
dim(X); qr(X)$rank
sqrt(mean(r^2)); max(abs(crossprod(X,r)))

rss <- sum(r^2)
s2.mle <- rss/n
s2.unbiased <- rss/(n-ncol(X))
ll <- function(b,s2) {
 e <- y-as.vector(X%*%b)
 -n/2*log(2*pi*s2)-sum(e^2)/(2*s2)
}
ex2 <- c(0,0,1,0,0,0)
tt <- seq(-1.25,1.25,length.out=251)
prof <- sapply(tt, function(t) ll(bhat+t*ex2,s2.mle))
tt[which.max(prof)]

f <- function(x) exp(1.5*x)-3*(x+6)^2-.05*x^3
g <- function(x) 1.5*exp(1.5*x)-6*(x+6)-.15*x^2
z <- lapply(c(-15,0), function(s)
  optim(s,f,g,method="BFGS"))
sapply(z, function(a)
  c(x=a$par, f=a$value, grad=abs(g(a$par)),
    converged=a$convergence==0))

set.seed(43202); u <- rnorm(n)
x6 <- x[,1]+1e-4*u; Xp <- cbind(X,x6)
ystar <- y+1e-3*u
bp <- lm.fit(Xp,y)$coef
bs <- lm.fit(Xp,ystar)$coef
fp <- as.vector(Xp%*%bp); fs <- as.vector(Xp%*%bs)
kappa2 <- function(A) {d<-svd(A)$d; max(d)/min(d)}
c(kappa2(X),kappa2(Xp))
bp[c(2,7)]; bs[c(2,7)]-bp[c(2,7)]
c(RMSE=sqrt(mean((fs-fp)^2)),max=max(abs(fs-fp)))

d <- read.csv("data/data-manipulation.csv")
dim(d); sapply(d,class); sum(duplicated(d$observation_id))
colSums(is.na(d))
a <- d[!is.na(d$response),]
a$feature_sum <- a$x1+a$x2
aggregate(cbind(response,feature_sum)~group,a,
        function(v)c(n=length(v),mean=mean(v)))
head(a[order(a$response,decreasing=TRUE),
  c("observation_id","group","response","feature_sum")],3)
stopifnot(nrow(a)==sum(!is.na(d$response)),
        !anyNA(a$response),
        !anyDuplicated(a$observation_id))
```

AI use disclosure: I used AI to confirm my answers and put them in a PDF-friendly format via Overleaf.